MODERN PHYSICS

Mock Test 1

Time: 50 minutes Maximum marks: 30

Instructions. This paper has 30 multiple-choice questions. Each question has exactly one correct option and carries 1 mark. Mark your answers on the OMR sheet.

Data. $h = 6.63 \times 10^{-34}\,\text{J s}$, $\hbar = 1.055 \times 10^{-34}\,\text{J s}$, $c = 3 \times 10^{8}\,\text{m/s}$, $hc = 1240\,\text{eV nm}$, $1\,\text{eV} = 1.6 \times 10^{-19}\,\text{J}$, $e = 1.6 \times 10^{-19}\,\text{C}$, $m_e = 9.11 \times 10^{-31}\,\text{kg}$, $m_p/m_e = 1836$, $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^{9}\,\text{N m}^2/\text{C}^2$, $R = 1.097 \times 10^{7}\,\text{m}^{-1}$, $a_0 = 0.53\,\text{Å}$, hydrogen ground-state energy $= -13.6\,\text{eV}$.

Q1. A plane electromagnetic wave travelling in vacuum has an electric field amplitude of 300 V/m. The amplitude of its magnetic field is

(A) $0.1\,\mu\text{T}$ (B) $1\,\mu\text{T}$ (C) $10\,\mu\text{T}$ (D) $100\,\mu\text{T}$

Q2. In Hertz's experiment, a spark appears across the small air gap of a receiving loop that is not electrically connected to the source. This confirms that

(A) changing electric and magnetic fields sustain each other and travel through empty space as an electromagnetic wave

(B) light is made of discrete quanta called photons

(C) electrons show wave-like diffraction

(D) a static electric charge can induce a current in a distant loop

Q3. Which of the following photoelectric-effect observations could **not** be explained by the classical wave picture of light?

(A) The photocurrent increases when a brighter light of suitable frequency is used.

(B) Light can transfer energy to electrons in a metal.

(C) Light of a given colour shows interference and diffraction.

(D) The maximum kinetic energy of the emitted electrons depends on the frequency of the light but not on its intensity.

Q4. Light of wavelength 310 nm falls on a metal of work function 2.5 eV. The stopping potential is

(A) 1.5 V (B) 2.5 V (C) 4.0 V (D) 6.5 V

Q5. Monochromatic light of fixed frequency above the threshold frequency falls on a metal surface. If the intensity of the light is doubled, then

(A) the maximum kinetic energy of the photoelectrons doubles and the saturation current is unchanged

(B) the maximum kinetic energy of the photoelectrons doubles and the saturation current doubles

(C) the maximum kinetic energy of the photoelectrons is unchanged and the saturation current doubles

(D) both the maximum kinetic energy and the saturation current are unchanged

Q6. Electrons accelerated through a potential difference of 31 kV strike the anode of an X-ray tube. The minimum wavelength of the emitted X-rays is

(A) 0.04 Å (B) 0.2 Å (C) 0.4 Å (D) 4.0 Å

Q7. Which statement about the X-ray spectrum from an X-ray tube is correct?

(A) The cutoff wavelength and the sharp lines both depend only on the target element.

(B) The cutoff wavelength depends on the tube voltage, while the positions of the sharp lines depend on the target element.

(C) The cutoff wavelength depends on the target element, while the positions of the sharp lines depend on the tube voltage.

(D) The continuous part of the spectrum is produced by transitions between inner atomic shells.

Q8. A crystal can diffract X-rays because

(A) X-rays have wavelengths of about 1 mm, comparable to the size of a crystal

(B) the spacing of crystal planes is about 1 μm, much larger than the X-ray wavelength

(C) X-ray photons carry charge and are deflected by the ions in the crystal

(D) the spacing between atomic planes (about an Ångström) is comparable to the X-ray wavelength, so waves scattered from different planes interfere

Q9. The de Broglie wavelength of an electron accelerated from rest through 150 V is approximately

(A) 0.5 Å (B) 1.0 Å (C) 1.4 Å (D) 2.0 Å

Q10. An electron and a proton have the same kinetic energy. The ratio λ_e/λ_p of their de Broglie wavelengths is approximately

(A) 1/1836 (B) 1/43 (C) 1 (D) 43

Q11. An electron is localized within a region of width $\Delta x = 1.0$ Å. The minimum uncertainty in its momentum allowed by the uncertainty principle is about

(A) 3.3×10^{-24} kg m/s (B) 1.1×10^{-24} kg m/s (C) 5.3×10^{-25} kg m/s (D) 2.6×10^{-25} kg m/s

Q12. A very weak X-ray beam passes through a narrow slit so that photons arrive at the screen one at a time. After a very large number of photons have been detected,

(A) each photon has produced one localized hit at an unpredictable position, and together the hits build up the diffraction pattern

(B) each photon has produced its own complete but faint diffraction pattern

(C) no pattern forms, because a single photon cannot interfere with other photons

(D) a pattern forms only if all the photons are sent through the slit at the same instant

Q13. A beam of hydrogen atoms in the ground state ($l = 0$) passes through a non-uniform magnetic field and splits into two beams. This is evidence that the electron

(A) has an intrinsic angular momentum (spin) with two possible projections, $m_s = \pm\frac{1}{2}$

(B) has orbital angular momentum $l = 1$ in the ground state

(C) has a different charge in each of the two beams

(D) occupies two different principal energy levels at the same time

Q14. The magnitude of the orbital angular momentum of an electron with $l = 2$ is

(A) $2\hbar$ (B) $\sqrt{6}\,\hbar$ (C) $3\hbar$ (D) $6\hbar$

Q15. Two electrons in the same atom are described by the quantum numbers (n, l, m_l, m_s). Which pair is **not** allowed?

(A) $(2,1,0,+\frac{1}{2})$ and $(2,1,0,-\frac{1}{2})$ (B) $(2,1,0,+\frac{1}{2})$ and $(2,1,1,+\frac{1}{2})$

(C) $(3,2,-1,-\frac{1}{2})$ and $(3,2,-1,-\frac{1}{2})$ (D) $(1,0,0,+\frac{1}{2})$ and $(2,0,0,+\frac{1}{2})$

Q16. Spin-orbit coupling splits the $2p$ level into $2p_{3/2}$ and $2p_{1/2}$. As a result, the K_α X-ray line (the $L \to K$ transition)

(A) remains a single line, because spin-orbit coupling affects only the $1s$ level

(B) disappears, because the split $2p$ levels can no longer decay to $1s$

(C) splits into three lines, one for each value of m_l of the $2p$ level

(D) splits into two closely spaced lines, $K_{\alpha 1}$ ($2p_{3/2} \to 1s_{1/2}$) and $K_{\alpha 2}$ ($2p_{1/2} \to 1s_{1/2}$)

Q17. In Rutherford's gold-foil experiment, a very small fraction of the alpha particles (about 2×10^{-4}) was scattered backwards. This showed that

(A) the positive charge and most of the mass of the atom are concentrated in a very small nucleus

(B) the positive charge is spread uniformly through the whole atom

(C) electrons carry most of the mass of the atom

(D) alpha particles are absorbed by atomic electrons and re-emitted

Q18. A 7.7 MeV alpha particle moving head-on towards a gold nucleus ($Z = 79$) has a distance of closest approach of 3.0×10^{-14} m. For the same alpha particle moving head-on towards a copper nucleus ($Z = 29$), the distance of closest approach is about

(A) 4.0×10^{-15} m (B) 1.1×10^{-14} m (C) 3.0×10^{-14} m (D) 8.2×10^{-14} m

Q19. The longest wavelength in the Balmer series of hydrogen is about

(A) 91.2 nm (B) 121.5 nm (C) 365 nm (D) 656 nm

Q20. A hydrogen-like ion has a single electron and nuclear charge $Z = 10$. Using $E_n = -13.6\,Z^2/n^2$ eV, the energy of the photon emitted when the electron falls from $n = 2$ to $n = 1$ is

(A) 10.2 eV (B) 102 eV (C) 1.02 keV (D) 1.36 keV

Q21. According to Bohr's correspondence principle, the predictions of quantum theory merge with those of classical physics when

(A) the electron is in its lowest state, $n = 1$

(B) the atomic number Z is very large

(C) the quantum numbers are very large, for example for transitions $n \to n - 1$ with $n \gg 1$

(D) Planck's constant is made larger

Q22. In a Franck–Hertz experiment with mercury vapour, the collector current drops sharply at accelerating voltages of 4.9 V, 9.8 V, 14.7 V, and so on. This shows that

(A) mercury atoms absorb energy from electrons only in discrete amounts (about 4.9 eV), indicating discrete atomic energy levels

(B) electrons lose energy continuously in elastic collisions with mercury atoms

(C) the filament emits electrons only at multiples of 4.9 V

(D) mercury atoms are ionised at 4.9 V

Q23. A particle has the wavefunction $\psi(x) = Ax$ for $0 \le x \le 1$ and $\psi(x) = 0$ elsewhere. The normalization constant A is

(A) $1/3$ (B) $1/\sqrt{3}$ (C) 1 (D) $\sqrt{3}$

Q24. The ground-state energy of a particle in a one-dimensional infinite square well of width L is E. If the width is doubled to $2L$, the ground-state energy becomes

(A) $E/8$ (B) $E/4$ (C) $E/2$ (D) $2E$

Q25. An electron in a one-dimensional infinite square well has ground-state energy $E_1 = 2.0$ eV. The energy of the photon emitted in the transition $n = 3 \to n = 2$ is

(A) 2.0 eV (B) 6.0 eV (C) 10 eV (D) 18 eV

Q26. Adjacent energy levels of a quantum harmonic oscillator are 0.20 eV apart. The energy of the first excited state ($n = 1$), measured from the bottom of the potential well, is

(A) 0.20 eV (B) 0.30 eV (C) 0.40 eV (D) 0.50 eV

Q27. Which statement about a particle of energy $E < U$ in a finite square well of depth U is correct?

(A) The wavefunction must vanish at the walls, exactly as in the infinite well.

(B) The energy of each state is higher than in an infinite well of the same width.

(C) The wavefunction and its derivative are discontinuous at the walls.

(D) The wavefunction decays exponentially outside the well, so there is a nonzero probability of finding the particle in the classically forbidden region.

Q28. For the hydrogen atom with principal quantum number $n = 3$, the allowed values of the orbital quantum number l are

(A) $0, 1, 2$ (B) $0, 1, 2, 3$ (C) $1, 2, 3$ (D) $0, 1$

Q29. The number of radial nodes (nodal spheres) in the $4p$ orbital of hydrogen is

(A) 1 (B) 2 (C) 3 (D) 4

Q30. For the ground state of hydrogen, the most probable distance of the electron from the nucleus is a_0. The average distance $\langle r \rangle$ is

(A) $0.5\,a_0$ (B) a_0 (C) $1.5\,a_0$ (D) $2\,a_0$

End of question paper

Answer Key and Solutions

1	2	3	4	5	6	7	8	9	10
B	A	D	A	C	C	B	D	B	D

11	12	13	14	15	16	17	18	19	20
C	A	A	B	C	D	A	B	D	C

21	22	23	24	25	26	27	28	29	30
C	A	D	B	C	B	D	A	B	C

Q1. (B) *Electromagnetic waves, Lecture 1-2*

In vacuum $|E| = c|B|$, so $B_0 = E_0/c = \dfrac{3 \times 10^2}{3 \times 10^8} = 10^{-6}$ T $= 1\,\mu$T.

Q2. (A) *Hertz experiment, Lecture 1-2*

The receiver has no wire connection to the source, so the energy must cross the gap through space. This is exactly Maxwell's prediction: a changing $\vec{E}$ makes a changing $\vec{B}$ and vice versa, and the pair propagates. Had Maxwell been wrong, there would be no spark.

Q3. (D) *Photoelectric effect, Lecture 1-2*

Options A, B and C are all consistent with a classical wave (more intensity means more energy delivered, light carries energy, light interferes). The wave picture predicts that a larger field amplitude (intensity) gives more kinetic energy and that any frequency should work eventually. Observation shows $K_{\max} = h\nu - \phi$, which depends on frequency only, with a threshold frequency.

Q4. (A) *Photoelectric effect, Lecture 1-2*

Photon energy $= hc/\lambda = 1240/310 = 4.0$ eV. Then $K_{\max} = 4.0 - 2.5 = 1.5$ eV and $eV_s = K_{\max}$ gives $V_s = 1.5$ V. The other options are the photon energy (4.0), the work function (2.5) and their sum (6.5).

Q5. (C) *Photoelectric effect, Lecture 1-2*

At fixed frequency, intensity is the number of photons per unit area per second. Doubling it doubles the number of electrons ejected, so the saturation current doubles. Each photon still has energy $h\nu$, so $K_{\max} = h\nu - \phi$ is unchanged.

Q6. (C) *X-ray production, Lecture 3*

The most energetic photon carries the whole electron energy: $eV = hc/\lambda_{\min}$. So $\lambda_{\min} = \dfrac{1240 \text{ eV nm}}{31000 \text{ eV}} = 0.040$ nm $= 0.4$ Å. Quick rule: $\lambda_{\min}(\text{Å}) \approx 12.4/V(\text{kV})$. As a check, 50 kV gives 0.25 Å, so a lower voltage must give a longer wavelength.

Q7. (B) *Bremsstrahlung and characteristic X-rays, Lectures 3 and 6*

The continuous part comes from electrons decelerated near nuclei, and its short-wavelength cutoff is set by the electron energy eV. The sharp lines come from filling inner-shell vacancies, so their energies are differences of atomic levels and are characteristic of the element.

Q8. (D) *X-ray diffraction, Lecture 3*

Diffraction is strong when the wavelength is comparable to the spacing of the scatterers. X-rays have $\lambda \sim 0.3$ Å and atomic planes in a crystal are separated by about an Ångström, so a crystal acts as a natural grating.

Q9. (B) *de Broglie wavelength, Lecture 4*

$\lambda = \dfrac{h}{\sqrt{2m_e eV}} = \dfrac{6.63 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(1.6 \times 10^{-19})(150)}} \approx 1.0 \times 10^{-10}$ m. Shortcut for electrons: $\lambda(\text{Å}) \approx \sqrt{150/V}$, which gives exactly 1 Å at 150 V.

Q10. (D) *de Broglie wavelength, Lecture 4*

At fixed kinetic energy K, $\lambda = h/\sqrt{2mK} \propto m^{-1/2}$. Hence $\lambda_e/\lambda_p = \sqrt{m_p/m_e} = \sqrt{1836} \approx 43$. The lighter particle has the longer wavelength.

Q11. (C) *Uncertainty principle, Lecture 4*

$\Delta p \geq \dfrac{\hbar}{2\Delta x} = \dfrac{1.055 \times 10^{-34}}{2 \times 10^{-10}} = 5.3 \times 10^{-25}$ kg m/s. The other options come from using $h/2\Delta x$ (3.3×10^{-24}), $\hbar/\Delta x$ (1.1×10^{-24}) or $\hbar/4\Delta x$ (2.6×10^{-25}).

Q12. (A) *Diffraction in the photon picture, Lectures 3 and 4*

Each detection is localized and its position cannot be predicted. The wave description fixes the probability of detection at each position, so many hits reproduce the wave-optics pattern. Nothing needs the photons to be present simultaneously.

Q13. (A) *Stern-Gerlach experiment and spin, Lecture 5*

With $l = 0$ there is no orbital moment and a single spot is expected. Two spots mean an intrinsic angular momentum with two projections, $2s + 1 = 2$, so $s = \frac{1}{2}$ and $m_s = \pm\frac{1}{2}$.

Q14. (B) *Orbital angular momentum, Lecture 5*

$L = \hbar\sqrt{l(l+1)} = \hbar\sqrt{6} \approx 2.45\hbar$. A quick check: L lies between $l\hbar$ and $(l+1)\hbar$.

Q15. (C) *Pauli exclusion principle, Lecture 6*

No two electrons in an atom can share the same complete set (n, l, m_l, m_s). In option C both electrons have identical sets. In A the spins differ, in B m_l differs, and in D n differs.

Q16. (D) *Spin-orbit coupling and X-ray fine structure, Lecture 6*

The $1s$ level ($l = 0$) is not split, but $2p$ splits into $2p_{3/2}$ and $2p_{1/2}$ with slightly different energies. Both decay to $1s_{1/2}$, giving two close lines, $K_{\alpha 1}$ and $K_{\alpha 2}$.

Q17. (A) *Rutherford scattering, Chapter 4*

A backward deflection needs a very strong force from something massive and charged that most alpha particles never come near. So the positive charge and mass are concentrated in a tiny nucleus, and the atom is mostly empty space.

Q18. (B) *Distance of closest approach, Chapter 4*

Head-on, all the kinetic energy becomes Coulomb energy: $K = \dfrac{1}{4\pi\varepsilon_0}\dfrac{2Ze^2}{R}$, so $R \propto Z$ at fixed K. Then $R_{\mathrm{Cu}} = 3.0 \times 10^{-14} \times \dfrac{29}{79} \approx 1.1 \times 10^{-14}$ m. The other options correspond to R unchanged, $R \propto 1/Z$ (8.2×10^{-14}) and $R \propto Z^2$ (4.0×10^{-15}).

Q19. (D) *Hydrogen spectral series, Chapter 4*

$\dfrac{1}{\lambda} = R\left(\dfrac{1}{2^2} - \dfrac{1}{3^2}\right) = \dfrac{5R}{36}$, so $\lambda = \dfrac{36}{5(1.097 \times 10^7)} = 6.56 \times 10^{-7}$ m $= 656$ nm. This is the smallest energy gap in the series, hence the longest wavelength. The Balmer limit ($n \to \infty$) is the shortest, $4/R = 365$ nm. The values 91.2 and 121.5 nm belong to the Lyman series.

Q20. (C) *Hydrogen-like energy levels, Chapter 4 and Lecture 6*

For hydrogen the $2 \to 1$ photon is $13.6(1 - \frac{1}{4}) = 10.2$ eV. Energies scale as $Z^2 = 100$, so $E = 1020$ eV $= 1.02$ keV. Option B results from scaling by Z instead of Z^2, and option D is the ground-state energy 13.6×100 rather than the transition energy.

Q21. (C) *Correspondence principle, Chapter 4*

For $n \to n - 1$ with large n, the photon energy is $hf \approx 27.2\ \mathrm{eV}/n^3$, and the photon frequency equals the classical orbital frequency of the electron. Neighbouring levels become nearly continuous, so quantum and classical descriptions agree.

Q22. (A) *Franck-Hertz experiment, Chapter 4*

An electron with kinetic energy near 4.9 eV excites a mercury atom in an inelastic collision and loses almost all its energy, so it cannot overcome the retarding potential and the current drops. At 9.8 and 14.7 V the same happens for two and three successive collisions. The equal spacing shows a single discrete excitation energy.

Q23. (D) *Normalization, Chapter 5*

$\displaystyle\int_0^1 A^2x^2\,dx = \frac{A^2}{3} = 1 \Rightarrow A = \sqrt{3}.$

Q24. (B) *Particle in a box, Chapter 5*

$E_n = \dfrac{n^2h^2}{8mL^2} \propto \dfrac{1}{L^2}$. Doubling L divides the energy by 4.

Q25. (C) *Particle in a box, Chapter 5*

$E_n = n^2 E_1$, so $E_3 - E_2 = (9-4)E_1 = 5E_1 = 10$ eV. The option 6 eV is $E_2 - E_1$ and 18 eV is E_3.

Q26. (B) *Harmonic oscillator, Chapter 5*

$E_n = \left(n + \frac{1}{2}\right)\hbar\omega$ with $\hbar\omega = 0.20$ eV. For $n = 1$: $E_1 = 1.5 \times 0.20 = 0.30$ eV. (The ground state is already at 0.10 eV, the zero-point energy.)

Q27. (D) *Finite square well, Chapter 5*

In the finite well the wavefunction and its slope are continuous at the walls, and outside it is e^{-Cx} with $C = \sqrt{2m(U-E)}/\hbar$. It does not vanish at the walls, and the energies are lower than in the infinite well (the wave has more room, so a longer wavelength).

Q28. (A) *Hydrogen quantum numbers, Chapter 6*

$l = 0, 1, \ldots, n-1$. For $n = 3$ this gives $l = 0, 1, 2$ (the $3s$, $3p$ and $3d$ states).

Q29. (B) *Nodes of hydrogen orbitals, Chapter 6*

Radial nodes $= n - l - 1 = 4 - 1 - 1 = 2$. The $4p$ orbital also has $l = 1$ angular node, and the total is $n - 1 = 3$.

Q30. (C) *Radial probability of the 1s state, Chapter 6*

The radial probability $P(r) \propto r^2 e^{-2r/a_0}$ peaks at a_0 but has a long tail towards large r, so the mean lies beyond the peak. Evaluating it gives $\langle r \rangle = \frac{3}{2}a_0$.